

PHY202: End-Semester Exam (part B)

Duration: 150 minutes; Maximum marks: 34

1. Imaginary part of an analytic function $f(z)$, with $z = x + iy$, is given by $(y \cos y + x \sin y)e^x$. Find the function $f(z)$.

[4]

2. Consider the function, $f(z) = \frac{\cos(2z)}{4z^4 + 5z^2 + 1}$ of complex variable z . Find the location of all the poles of this function. Make use of the Residue theorem to evaluate the integral, $I = \int_{-\infty}^{\infty} f(x)dx$.

[3+3]

3. (a) For a positive real number a , find the Fourier transform of the function,

$$f(x) = \begin{cases} 1/a, & |x| < a \\ 0, & |x| > a \end{cases}$$

- (b) Use the answer to find the Fourier transform of the Dirac-delta function $\delta(x)$.

- (c) Evaluate the integral, $I = \int_{-\infty}^{\infty} \frac{\sin^2(x)}{x^2} dx$.

[3+2+2]

4. Consider the matrix $M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

- (a) Find the eigenvalues and eigenvectors of M .

- (b) Find the matrix $\sin^2(\pi M/3)$.

[3+4]

5. (i) Solve the differential equation $(y - x)y' + 2x + 3y = 0$.

- (ii) Find the particular solution of the differential equation, $y''' - y'' = 12x - 2$.

- (iii) Find the general solution of $y'' - 3y' - 4y = 3e^{2x} + 2\sin x$.

[3+3+4]